Q.1 Which of the following system of equation is diagonally dominant

A. $${array}{l} x + 8y + 5z = 2\ + y + 9z = 3\\11x + y - z = 4 {array}$$

B. $${array}{l} x + y + 9z = 3\ + 8y + 5z = 2\\11x + y - z = 4 {array}$$

C. $${array}{l} 11x + y - z = 4\ + y + 9z = 3\ + 8y + 5z = 2 {array}$$

D. $${array}{l} 11x + y - z = 4\ + 8y + 5z = 2\ + y + 9z = 3 {array}$$

Q.2 Full pivoting, in fact, is more .....than the partial pivoting.

A. Easiest

B. Complicated

Q.3 While solving the system of linear equations; x-y=2, -x+y=3 by Jacobi's method, if (0,0) be its first approximate solution, then which of the following is second approximate solution?

A. (2,3)

B. (2,0)

C. Second approximate solution doesn't exist as system is singular

D. (0,3)

Q.4 In full pivoting we interchange rows and columns such that the.....element in the matrix of the variables also get changed.

A. Middle

B. Smallest

C. Largest

D. None of the given choices

Q.5 Which of the following system of equation is diagonally dominant?

A. $${array}{*{20}{l}} {2x + 13y + 5z = 7} \\\ {8x + y - z = 5} \\\ {x + 2y + 9z = 3} {array}$$

B. $${array}{*{20}{l}} {x + 2y + 9z = 3} \\\ {8x + y - z = 5} \\\ {2x + 13y + 5z = 7} {array}$$

C. $${array}{*{20}{l}} {8x + y - z = 5} \\\ {x + 2y + 9z = 3} \\\ {2x + 13y + 5z = 7} {array}$$

D. $${array}{*{20}{l}} {8x + y - z = 5} \\\ {2x + 13y + 5z = 7} \\\ {x + 2y + 9z = 3} {array}$$

Q.6 While solving a system of linear equations by Gauss Jordon Method, after all the elementary row operations if there lefts also zeros on the main diagonal then which of the is true about the system?

A. System may have unique solutions

B. System may have multiple numbers of finite solutions

C. System may have infinite many solutions

D. System has no solution

Q.7 If system of equations is inconsistent then its means that it has .....

A. None of the given choices

B. Many solutions

C. No Solutions

D. Infinite Many solutions

Q.8 While using the Gauss-Seidel Method for finding the solution of the system of equation, the following system $$\begin{array}{l} x + 2y + 2z = 3\\x + 3y + 3z = 2\\x + y + 5z = 2 \end{array}$$ can be rewritten as

A. $${array}{l} x = {3}{2} - y - z\ = 2 - {x}{3} - z\ = {2}{5} - {x}{5} - {y}{5} {array}$$

B. $${array}{l} x = 3 - 2y - 2z\ = 2 - x - 3z\ = 2 - y - x {array}$$

C. $${array}{l} x = 3 - 2y - 2z\ = {2}{3} - {x}{3} - z\ = {2}{5} - {x}{5} - {y}{5} {array}$$

D. $${array}{l} x = 1 - {2}{3}y - {2}{3}z\ = 1 - {3}{2}x - {3}{2}z\ = 1 - {y}{2} - {x}{2} {array}$$

Q.9 While solving the system; x+4y = 4, 2x–y = 1 by Gauss-Seidel method, which of the following ordering will lead more rapidly to the desired order of accuracy?

A. x+4y =1, 2x- y =4

B. 2x-y = 1, x+4y = 4

C. x-2y = 1, x+4y =4

D. no need to reordering

Q.10 The linear equation 2x+0y-2=0 has----- solution(s).

A. unique

B. infinite

C. finite

D. no solution
